

E-COMMERCE SALES INSIGHTS DASHBOARD

PROJECT REPORT

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Project Domain:

Data Analytics | Business Intelligence | Power BI

Tools Used:

- Power BI Desktop
 - Microsoft Excel
 - Power Query
 - DAX
 - GitHub
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ABSTRACT

The E-Commerce Sales Insights Dashboard is a Business Intelligence solution designed to help organizations monitor sales performance, customer behavior, product profitability, and delivery operations. The project transforms raw transactional data into meaningful insights using Power BI dashboards and interactive visualizations.

The dashboard provides decision-makers with a comprehensive view of revenue trends, profit generation, customer segmentation, product performance, and logistics efficiency. Through data-driven insights, businesses can improve operational efficiency, increase profitability, and enhance customer satisfaction.

INTRODUCTION

The rapid growth of e-commerce has generated large volumes of transactional data. Organizations require efficient analytical tools to convert this data into actionable business insights.

This project aims to develop an interactive dashboard that enables stakeholders to:

- Track business performance
- Monitor sales growth
- Evaluate customer purchasing patterns

- Identify profitable products
- Assess delivery efficiency
- Support strategic decision-making

The dashboard acts as a centralized platform where management can quickly analyze key business metrics.

PROBLEM STATEMENT

E-commerce companies often struggle to monitor multiple business functions simultaneously.

Common challenges include:

- Difficulty identifying top-performing products
- Lack of visibility into customer purchasing behavior
- Poor understanding of regional sales performance
- Limited tracking of delivery operations
- Challenges in measuring profitability

To address these issues, a comprehensive dashboard is required to provide real-time insights and improve business performance.

PROJECT OBJECTIVES

The primary objectives of this project are:

Sales Objectives

- Analyze revenue trends
- Track sales growth
- Monitor category performance

Customer Objectives

- Identify high-value customers
- Analyze customer segments
- Measure customer retention

Product Objectives

- Evaluate product profitability
- Identify best-selling products
- Monitor category contribution

Delivery Objectives

- Track delivery performance
 - Analyze return rates
 - Improve logistics efficiency
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DATASET DESCRIPTION

The dataset consists of order-level e-commerce transactions.

Attributes Included

Field Name	Description
Order ID	Unique order identifier
Customer ID	Unique customer identifier
Customer Name	Customer name
Order Date	Date of purchase
Region	Sales region
State	Customer state
City	Customer city
Product Category	Product classification
Product Name	Product description
Quantity Sold	Units sold
Unit Price	Selling price
Discount %	Discount applied
Sales Amount	Total revenue
Cost Amount	Product cost
Profit Amount	Generated profit
Profit Margin %	Profit percentage
Payment Method	Mode of payment
Shipping Mode	Shipping type
Delivery Status	Order status

Field Name	Description
Customer Segment	Customer classification

DATA PREPROCESSING

Before analysis, the dataset underwent multiple preprocessing steps.

Data Cleaning Activities

Missing Value Treatment

- Identified null values
- Replaced missing entries
- Validated records

Duplicate Removal

- Removed duplicate transactions
- Ensured data consistency

Data Formatting

- Standardized date formats
- Corrected categorical values
- Formatted numerical fields

Data Validation

- Checked profit calculations
 - Validated sales totals
 - Verified quantity values
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EXPLORATORY DATA ANALYSIS

EDA was performed to understand data distribution and identify patterns.

Analysis Conducted

- Sales trend analysis
- Customer segmentation
- Product contribution analysis
- Regional performance analysis

- Delivery performance evaluation
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KEY PERFORMANCE INDICATORS (KPIs)

The dashboard includes the following business KPIs:

Revenue KPIs

- Total Sales
- Revenue Growth %
- Average Order Value

Profitability KPIs

- Total Profit
- Profit Margin %

Customer KPIs

- Total Customers
- Active Customers
- Repeat Customers
- Customer Lifetime Value

Delivery KPIs

- Delivered Orders
 - Return Rate
 - Delivery Success Rate
 - Average Delivery Time
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DASHBOARD DEVELOPMENT

The dashboard was developed using Power BI.

Dashboard Pages

1. Executive Overview

Provides a summary of overall business performance.

Visuals:

- KPI Cards
 - Sales Trends
 - Category Analysis
 - Regional Performance
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2. Sales Analysis

Focus Areas:

- Monthly Revenue Trends
- Revenue Growth
- Regional Sales
- Product Revenue

Visuals:

- Line Charts
 - Bar Charts
 - KPI Cards
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3. Product Analysis

Focus Areas:

- Best-Selling Products
- Product Profitability
- Category Contribution

Visuals:

- Treemap
 - Bar Charts
 - Product Rankings
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4. Customer Analysis

Focus Areas:

- Customer Growth
- Customer Segmentation
- Revenue Contribution
- Customer Profitability

Visuals:

- Donut Charts
 - Scatter Plots
 - Customer Ranking Charts
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5. Delivery Analysis

Focus Areas:

- Delivery Success Rate
- Return Analysis
- Shipping Performance
- Regional Delivery Performance

Visuals:

- Maps
 - Trend Charts
 - Status Distribution Charts
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BUSINESS INSIGHTS

Sales Insights

- Revenue shows consistent growth across periods.
- Certain categories contribute significantly to total sales.
- Regional differences indicate expansion opportunities.

Customer Insights

- Repeat customers generate substantial revenue.
- Customer retention positively impacts profitability.
- High-value customers contribute disproportionately to business growth.

Product Insights

- Electronics and premium products generate high revenue.
- High-margin products significantly improve profitability.

Delivery Insights

- Standard shipping accounts for most deliveries.

- Return rates impact operational costs.
 - Delivery efficiency influences customer satisfaction.
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BUSINESS RECOMMENDATIONS

Customer Recommendations

- Introduce loyalty programs
- Improve customer retention strategies
- Personalize marketing campaigns

Product Recommendations

- Focus on high-margin products
- Optimize inventory planning
- Monitor product profitability

Delivery Recommendations

- Reduce delivery delays
- Improve logistics tracking
- Minimize return rates

Sales Recommendations

- Expand high-performing categories
 - Explore growth opportunities in emerging regions
 - Optimize discount strategies
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PROJECT OUTCOME

The dashboard successfully transforms raw sales data into actionable business intelligence.

Benefits achieved:

- Faster decision-making
 - Better customer understanding
 - Improved operational visibility
 - Enhanced profitability tracking
 - Comprehensive performance monitoring
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FUTURE ENHANCEMENTS

Future improvements may include:

- Sales Forecasting using Machine Learning
 - Customer Churn Prediction
 - Inventory Optimization Models
 - Real-Time Data Integration
 - AI-Based Product Recommendation System
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CONCLUSION

The E-Commerce Sales Insights Dashboard provides a complete analytical framework for monitoring business performance. By integrating sales, customer, product, and delivery analytics into a single platform, the dashboard enables organizations to make informed decisions and achieve sustainable business growth.

The project demonstrates practical applications of Data Analytics, Business Intelligence, and Power BI in solving real-world business challenges.